

The Mathematical Structure of Rainfall Representations

1. A Review of the Stochastic Rainfall Models

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This is the first of a three-part series on the mathematical structure of rainfall models. Several important attempts at modeling rainfall are reviewed. Special attention is given to the mathematical structures that arise in the rainfall descriptions. A general overview of the three-part series is given as preface to this part.

THE SCOPE AND THE ORGANIZATION OF THE SERIES

The newly emerging areas of physically based statistical hydrology have created a need for exploring new mathematical tools applicable to the analysis of hydrologic processes from a physical viewpoint. Some examples of these areas include the theory of solute and water transport in porous media, sediment transport, the rainfall process, the rainfall-runoff process, and the streamflow process. Within the context of the rainfall process and other hydrologic processes 'driven by' rainfall inputs the developing role of the mathematical theory of point processes and random measures is noteworthy. Unfortunately the mathematics literature does not seem to contain a current expository treatment of the theory of point processes which is immediately accessible to an applied audience. Because of the void between the current mathematical developments of the point process theory and its practicality in hydrology, we consider it to be worthwhile that this area of mathematics be exposed to the hydrologic community with a demonstration of how it fits into the analysis of the structure of rainfall and rainfall based processes (e.g., streamflows). Although the title of this article is 'The Mathematical Structure of Rainfall Representations,' the scope of this article is not confined to rainfall alone. This will be clear to the reader from the organization of the paper. This series is divided into three parts: (1) A review of the stochastic rainfall models, (2) a review of the theory of point processes, and (3) some applications of the point process theory to rainfall processes.

In the first part we will present the key mathematical ideas (in our judgment) that have been introduced within the last two decades or so in modeling some of the empirically observed features of rainfall. This presentation is certainly not completely comprehensive nor does it include references to all the papers that have been published on this subject. The limitations of space and our limited competence preclude such an undertaking. For example, part 1 does not go deeply into the meteorological literature dealing with precipitation. Similarly, the second-order time series analysis of rainfall is not reviewed.

In the second part the theory of point processes is unveiled. This part can be read independently of the first part. Once again, in this part we sketch out the main mathematical ideas which are central to the development of a solid working

knowledge of the point process theory. The notion of a probability generating functional (pgfl) as a natural generalization of the concept of a probability generating function (pgf) plays a central role in the theory of point processes. We have tried to unfold the theory of point processes in such a way that its relevance toward various applications will be clear to the reader, although only the mathematical techniques are emphasized in this part. The third section of this part concerns operations on point processes which are particularly germane to the study of rainfall and other hydrologic processes derived from rainfall (e.g., streamflows, ground water recharge, etc.).

In the third part we demonstrate applications of the point process theory to the modeling of a variety of hydrologic processes. For example, some of the rainfall models reviewed in part 1 are cast into the point process mold. The reader will see that this setting increases the analytic capabilities for a further in-depth study of these processes. In addition to the rainfall models, the examples of streamflows and ground water level fluctuations are used to show how the smoothing operation on point processes enables one to couple a deterministic response function with a statistical input description. This coupling provides the complete stochastic description of the output process (in the form of its pgfl) as opposed to its second order structure only. The necessity of requirements beyond the second order analysis stems from the fact that typically these hydrologic processes are non-Gaussian. The power and scope of the mathematical techniques exposed in part 2 to modeling hydrologic processes should become more apparent to the reader in this part. However, our intent is only to demonstrate an application of the mathematical ideas developed in part 2 to various hydrologic processes. No part of this paper is all-exhaustive, but if it contributes to a basis for longer strides in the development of hydrologic theory, then our goal in writing this series will be accomplished.

INTRODUCTION

Rainfall modeling represents an area which cuts across the boundaries of hydrology. Meteorology, atmospheric physics, climatology, and hydrology represent some of the disciplines which encompass one aspect or another of the rainfall phenomenon. In view of the breadth of scope occupied by rainfall analysis, it would be presumptuous on our part to attempt to review it entirely in one article. In selecting material for the review we were guided by a search for the most basic in-

gradients of known rainfall descriptions. Rather than fitting data to specific distributions, piecing together the resulting descriptions, and then coming up with a model that 'happens to work,' we prefer the point of view that some basic physical (observed) structures be identified and then represented mathematically so that the source of the results can be understood. The data check then follows as the final step. (See *Aris* [1979] for a very nice treatment of the ideas underlying model building.)

One aspect of rainfall with which hydrologists have been most concerned is modeling its statistical structure. A closer look at the literature along these lines reveals that although many mathematical models have been proposed on the structure of rainfall, there is no unified mathematical approach to modeling rainfall. In part this difficulty stems from the nature of rainfall itself. The structure of rainfall in different parts of the globe, or even different parts of a large country such as the United States, exhibits considerable variability. Therefore, the development of a single 'model' that incorporates all of these variabilities would indeed be a hopeless task. However, another part of this difficulty is due to the limitations on the availability of 'appropriate' mathematical tools designed to exploit the dependence which the rainfall phenomenon seems to exhibit (i.e., the clustering dependence). The more readily available tools (e.g., the theory of Markov chains, renewal processes, Poisson processes, and other processes with independent increments), do not provide enough flexibility to study the cluster dependence shown by rainfall.

Since rainfall is a space-time phenomenon, its representation in space and time requires concepts from the theory of random fields. Although the Poisson process extends to Poisson fields, it is not always easy to extend the mathematical ideas underlying stochastic processes to random fields. For example, the theory of Markov processes is based on the natural ordering of the numbers on the real line, making it possible to define the past, present, and the future. In higher dimensional space this ordering is lost representing nontrivial difficulties in generalizing the Markov process theory to Markov random fields [see *Dynkin*, 1980]. Fortunately, the application of Markov random fields to modeling space-time rainfall does not seem natural because the cluster dependence exhibited by space-time rainfall can be handled more directly within the framework of cluster point processes. Moreover, in the realm of the commonly available probabilistic tools it indeed seems to be a Herculean task to investigate the structure of other processes which are 'driven by rainfall' (e.g., streamflows, subsurface flows, sediment yields, etc.). Therefore it is not surprising that even in the newly emerging field of the stochastic modeling of streamflows from a physical basis [*Klemeš*, 1978], the rainfall structure is treated rather casually from a physical viewpoint and the analysis of the physically based processes are generally restricted to obtaining their second-order structure.

Although brief, the remarks made above indicate a few of the obvious difficulties which hydrologists have faced in the last two decades in being able to design a unified mathematical approach to modeling rainfall. The purpose of our review in this part is to expand on these remarks by reviewing the main mathematical ideas that have been introduced into modeling rainfall. In this sense we did not consider it necessary to include the entire spectrum of papers that have been written regarding rainfall analysis, so only 'key' papers and some earlier reviews [see, e.g., *Kavvas and Delleur*, 1975; *Todorovic and*

Woolhiser, 1976] were selected for describing the state of the art. A comprehensive list of papers on rainfall written over the past half decade has been compiled by *Court* [1979].

In this paper we have not reviewed the second-order models on rainfall (e.g., autoregressive, ARMA, etc.). On one hand, these models have been employed to model rainfall on large time scales (e.g., monthly sequences) whereas the present review focuses on rainfall modeling at short time scales (e.g., hours, or at the most, days) for which the second-order models are not appropriate. On the other hand, the second-order models deal only with the covariance structure of a stochastic process. Although covariance is an important measure of dependence, in general it does not capture the complete dependence structure of a stochastic process. Therefore features such as the maximum rainfall per storm, the sequences of periods with no rainfall, and other crossing properties derived from rainfall, etc., cannot be investigated from a second-order model.

An important research area of rainfall analysis consists of investigating the long-term structure of rainfall. Most of the research in this domain has been directed toward the nature of dependencies within rainfall sequences (e.g., annual rainfall) for exhibiting Hurst phenomenon. Attempts have not been made to investigate any nonstationarities (beyond the annual cyclicity) that may be present in the rainfall structure [see, e.g., *Potter*, 1979]. The present review does not include this aspect of the rainfall phenomenon. We think that investigations on the long-term behavior of rainfall merit a separate review.

The organization of this part of the paper is summarized as follows: (1) The temporal rainfall process, including counting processes, rainfall depth and duration processes, and some results, (2) the structure of rainfall in space and time including a summary of observed regularities (clustering) in subsynoptic space-time rainfall and mathematical structure of space-time rainfall, and (3) a perspective on the directions for further research.

THE TEMPORAL STRUCTURE OF RAINFALL

Two sequences of random variables enter very naturally into the description of the temporal structure of rainfall. The first sequence counts the number of 'rainfall events' within various intervals of time. The second sequence associates certain rainfall amounts with the rainfall events. The definition of a rainfall event depends on the time scale over which the rainfall process is to be described. Both discrete and continuous time scales have been used in the literature. We begin this discussion with a review of the counting processes used to describe the occurrence of rainfall events. This will then be followed by a review of the models for rainfall magnitudes per rainfall event.

The counting process. Let us consider a discrete time scale of, say, days. Let $X_1, X_2, \dots, X_n, \dots$ be a sequence of random variables such that X_i denotes the rainfall amount on the i th day. Define a new sequence $\{Y_i\}$, as

$$\begin{aligned} Y_i &= 1 & X_i > 0 \\ Y_i &= 0 & X_i = 0 \end{aligned} \quad (1)$$

and $N_n = N(0, n) = \sum_{i=1}^n Y_i$. Then, for an arbitrary but fixed time interval $(0, n)$, the variable N_n counts the number of days in $(0, n)$ for which $Y_i = 1$. In order to specify the probabilistic structure of N_n we begin with the simplest assumption on the

sequence $\{Y_i\}$. Namely, consider the case in which the variables Y_i are independent and identically distributed (iid) with $P(Y_i = 1) = p$ and $P(Y_i = 0) = q = 1 - p$ $i = 1, 2, \dots$. Then it follows easily that N_n has a binomial density

$$P(N_n = k) = \binom{n}{k} p^k q^{n-k} \quad 0 \leq k \leq n \quad (2)$$

In order to completely specify the stochastic process $\{N_n\}$, it is necessary that all of its finite dimensional joint distribution (density) functions be determined [Karlin and Taylor, 1975]. This is straight forward in the above case because the increments $N_n - N_{n-1}$ over disjoint intervals (n_{i-1}, n_i) , $i = 1, 2, \dots$, are independent and in each interval the density of $(N_n - N_{n-1})$ is known (binomial). For an application of this process to daily rainfall we refer to Smith and Schreiber [1973].

Although the binomial process has not been used extensively in the literature, its analogue in the continuous time scale, the Poisson process, has been used extensively. Consider a continuous time interval $(0, t)$ and discretize this interval into n subintervals of length Δt such that $n = [t/\Delta t]$, where $[\]$ denotes the largest integer smaller than $t/\Delta t$. Define the binomial process over the discretized time scale. Then, as $n \rightarrow \infty$, $p \rightarrow 0$ such that $np \rightarrow \lambda t$, (2) converges to a Poisson distribution [Feller, 1968, p. 158]. Let $N(t) \equiv N(0, t)$ represent the number of rainfall events (to be explained later) that occur in $(0, t)$. Then the density of $N(t)$ is given by

$$P(N(t) = k) = e^{-\lambda t} (\lambda t)^k / k! \quad k = 0, 1, \dots \quad (3)$$

Because of the independence of the increments $(N_n - N_{n-1})$ over disjoint time intervals (n_{i-1}, n_i) , $i = 1, 2, \dots$, the independence of $N(t) - N(t_{i-1})$ over disjoint intervals (t_{i-1}, t_i) , $i = 1, 2, \dots$, is retained and a specification of the finite dimensional joint distributions of the Poisson process $\{N(t)\}$ is determined. The parameter λ can be viewed as the rate at which the events occur in time. A variant of (3) is obtained by allowing the parameter $\lambda(s)$, $0 < s < \infty$, to be a time dependent rate function. In this case (3) is modified according to

$$P(N(t) = k) = e^{-\Lambda(t)} \Lambda(t)^k / k! \quad k = 0, 1, \dots$$

with

$$\Lambda(t) = \int_0^t \lambda(s) ds \quad (4)$$

This process is often called a nonhomogeneous Poisson process. For an application of the Poisson process to temporal rainfall we refer to Shane [1964], Todorovic and Yevjevich [1969], Gupta and Duckstein [1975], Todorovic and Woolhiser [1976], and Eagleson [1978].

It may be noticed that in passing from the discrete to the continuous time scale the definition of a rainfall event becomes obscure, since a rainfall event does not occur instantaneously in time. In order to circumvent this difficulty at least partially, two types of assumptions have been introduced. According to the first assumption, one assumes a rainfall event to be an instantaneous entity and then restricts the description to a particular class of storms, e.g., the air mass thunderstorms [Gupta and Duckstein, 1975]. In the other case the number of termination epochs of rainfall bursts (events) within the time interval $(0, t)$ are counted [see Todorovic and Yevjevich, 1969].

A serious drawback in the specification of the above schemes, whether in discrete time or in continuous time, is the

assumption that the increments of the counting process over disjoint time intervals are independent. Since this assumption cannot be retained, in general [see, e.g., LeCam, 1961; Smith and Schreiber, 1973; Kavvas and Delleur, 1975; and Todorovic and Woolhiser, 1976], one is led to models where these increments are stochastically dependent. The most popular modification of the assumption of the independence among the variables, Y_n in (1) is to assume them to be governed by a discrete parameter Markov chain. The probabilistic structure of a Markov chain is completely determined by a transition probability matrix and an initial probability distribution [see, e.g., Feller, 1968]. In the present context, assuming the variable Y_i to be Markov dependent, let the transition matrix be denoted by

$$P = \begin{bmatrix} p_0 & q_0 \\ p_1 & q_1 \end{bmatrix} \quad q_0 = 1 - p_0 \quad p_1 = 1 - q$$

where

$$\begin{aligned} p_0 &= P(Y_i = 0 | Y_{i-1} = 0) & i = 1, 2, \dots \\ p_1 &= P(Y_i = 0 | Y_{i-1} = 1) & i = 1, 2, \dots \end{aligned} \quad (5)$$

In addition, let $\pi_0 = P(Y_0 = 0)$ and $\pi_1 = P(Y_0 = 1)$ denote the respective elements of the initial distribution. Defining N_n as before, it can be shown that

$$P(N_n = k) = \pi_1 \psi_1(k, n) + \pi_0 \psi_0(k, n) \quad (6)$$

where

$$\begin{aligned} \psi_0(k, n) &= q_1^k p_0^{n-k} + \sum_{c=1}^{\infty} \binom{k-1}{b-1} \binom{n-k}{a} \left(\frac{p_1}{p_0} \right)^a \left(\frac{q_0}{q_1} \right)^b \\ c_0 &= n + \frac{1}{2} - |2k - n - \frac{1}{2}| \quad k > 0 \\ c_0 &= 0 \quad k = 0 \\ a &= \min \{v: v \geq \frac{1}{2}(c-1)\} \\ b &= \min \{v: v > \frac{1}{2}c\} \end{aligned} \quad (7)$$

Similarly,

$$\begin{aligned} \psi_1(k, n) &= q_1^k p_0^{n-k} + \sum_{c=1}^{\infty} \binom{k}{a} \binom{n-k-1}{b-1} \left(\frac{p_1}{p_0} \right)^a \left(\frac{q_0}{q_1} \right)^b \\ c_1 &= n + \frac{1}{2} - |2k - n + \frac{1}{2}| \quad k < n \\ c_1 &= 0 \quad k = n \end{aligned} \quad (8)$$

and a, b remain the same as given in (7). This result was first obtained by Gabriel [1959]. Subsequently, it has been applied to daily rainfall by, among others, Gabriel and Neumann [1962], Smith and Schreiber [1963], and Todorovic and Woolhiser [1976]. The above description refers to a homogeneous Markov chain in so far as the transition matrix P remains time invariant. A natural variant of the above structure is obtained by allowing P to vary with time due to 'seasonality.' This results in a nonhomogeneous Markov chain and has been applied to daily rainfall by Caskey [1963] and Feyerherm et al. [1965], for example. The reader will notice that the above description of the stochastic counting process $\{N_n; n \geq 1\}$ only gives its marginal probability density. Hence it does not represent a complete specification of its stochastic structure (i.e., all of its finite dimensional density functions). In particular it should be noticed that although the sequence $Y_1, Y_2, \dots$ is Markov dependent, the same will not generally be true for the

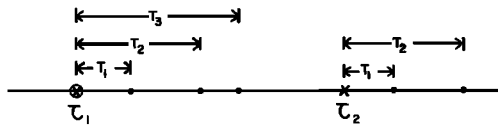


Fig. 1. A schematic depiction of the Neymann-Scott process.

sequence $N_1, N_2, \dots$. To see the source of the extra dependence, simply note that for $r_1 < r_2 < \dots < r_k < n$,

$$\begin{aligned} P(N_n = l | N_{r_1}, \dots, N_{r_k}) &= P\left(\sum_{i=r_k+1}^n Y_i = l - \sum_{i=1}^{r_k} Y_i | N_{r_1}, \dots, N_{r_k}\right) \\ &= E\left[P\left(\sum_{i=r_k+1}^n Y_i = l - N_{r_k} | Y_{r_k}, N_{r_k}\right)\right] \\ &= E[P(N_n = l | Y_{r_k}, N_{r_k})] \end{aligned}$$

where $E[\]$ denotes the expectation (with respect to Y_{r_k}). This shows that the future evolution of $\{N_i\}$, given its present and the past states, need not depend only on its present state as it should in order to be Markovian. However, the reader may note that $\{N_n, Y_i\}$ are jointly Markovian.

In extending the above picture to a continuous time scale, two different schemes have been introduced into the literature. In the first scheme one assumes that $\{Y(t); 0 \leq t < \infty\}$ is a continuous parameter, two-state Markov chain. The random variables $\{Y(t)\}$ alternate between 0 and 1 with the former denoting 'no rainfall' and the latter denoting 'nonzero rainfall' [Crovelli, 1971]. In the second scheme the counting process $\{N(t); 0 \leq t < \infty\}$, denoting the number of rainfall events, is assumed to be a nonhomogeneous pure birth Markov process [see, e.g., Todorovic and Yevjevich, 1969; Todorovic and Woolhiser, 1976]. For the present we will consider only the counting process $\{N(t)\}$ with a deferment of our discussion regarding the process $\{Y(t)\}$ to the review of models on rainfall magnitudes.

Since $\{N(t)\}$ is postulated to be a nonhomogeneous pure birth Markov process, let $P(s, t) = [p_{ij}(s, t)]$, $0 < s < t$, denote its transition probability (function) matrix:

$$p_{ij}(s, t) = P(N(t) = j | N(s) = i) \quad i, j = 0, 1, 2, \dots \quad (9)$$

and let $\pi_i = P(N(0) = i)$ be its initial density function. In order to obtain $P_{ij}(t) = P_{ij}(0, t)$, one begins with a specification of the infinitesimal generator (matrix), $Q(t) = [q_{ij}(t)]$, as follows [see, e.g., Feller, 1968; Karlin and Taylor, 1975]:

$$q_{ij}(t) = \lim_{s \downarrow 0} \frac{p_{ij}(t, t+s) - p_{ij}(t, t)}{s} \quad (10)$$

The matrix Q satisfies

$$\sum_j q_{ij}(t) = 0 \quad q_{ii}(t) < 0 \quad q_{i,i+1}(t) > 0 \quad (11)$$

By the assumption that $\{N(t)\}$ is a pure birth process, Q also satisfies,

$$q_{ij}(t) = 0 \quad j > i + 1 \quad \text{or} \quad j < i \quad (12)$$

In view of (11) and (12) we have $-q_{ii}(t) = q_{i,i+1}(t) = \lambda_i(t)$, say, in the notation of Todorovic and Woolhiser [1976]. Now, (10), (11), and (12), in conjunction with a formal differentiation of the Chapman-Kolmogorov equation, yield the Kolmogorov's forward equation for $p_{ij}(t)$:

$$p'_{ij}(t) = -\lambda_i(t)p_{ij}(t) + \lambda_{j-1}(t)p_{i,j-1}(t) \quad (13)$$

In (13), $p'_{ij}(t)$ denotes the derivative of $p_{ij}(t)$. By summing both sides of (13) with respect to the initial density π_i , it follows that the marginal density of $N(t)$

$$p_j(t) = \sum_i \pi_i p_{ij}(t) \quad (14)$$

also satisfies the forward equation. This equation, given by

$$p'_j(t) = -\lambda_j(t)p_j(t) + \lambda_{j-1}(t)p_{j-1}(t) \quad (15)$$

represents the form in which Todorovic and Woolhiser [1976, pp. 15–22] write their differential equation. In the hydrologic literature, many variants of (15) have been introduced in connection with the temporal rainfall process. These variations are obtained by imposing particular assumptions on the intensity $\lambda_j(t)$. In one form or another, however, the processes have been known in the probability literature for quite some time (e.g., as classical models of population growth).

To begin with, let us assume that $\lambda_j(t) \equiv \lambda(t)$. In this case the solution of (15) leads to (4), i.e., $\{N(t)\}$ becomes a nonhomogeneous Poisson process. This was discussed earlier. If we set $\lambda_j(t) = (a + j)b(t)$, where $a > 0$ is a real number, then the solution of (15), under the initial condition $p_0(0) = 1$, becomes

$$\begin{aligned} p_j(t) &= \frac{\Gamma(a+j)}{\Gamma(a)j!} (e^{-B(t)})^a (1 - e^{-B(t)})^j \\ B(t) &= \int_0^t b(s) ds \quad j = 0, 1, 2, \dots \end{aligned} \quad (16)$$

Equation (16) is a time-dependent version of the Polya contagion process [Todorovic and Woolhiser, 1976]. In particular, if $b(t) = 1/(a + t)$, then $B(t) = \log a^{-1}(a + t)$. In this case (16) reduces to the classical Polya process [see Feller, 1968, p. 480]. In the same manner the Yule process can be obtained from (16) by choosing $b(t) = \lambda$ [see Feller, 1968, p. 450].

The Polya urn scheme was introduced by Wiser [1965] for counting the number of rainfall events. The continuous time version of the Polya process is easily derived from the Polya urn scheme [see Feller, 1968, p. 480]. Wiser [1965] also introduced other urn models for modeling precipitation events; however, a discussion of all of these models is beyond the scope of this paper.

Motivated by the work of Wiser, Kavvas and Delleur [1975] introduced the Neyman-Scott (N-S) cluster process for modeling rainfall events. The structure can be understood as follows. Observations suggest that one can think of the occurrence of rainfall events in the form of clusters as being triggered by some generating mechanisms, e.g., storm fronts sweeping over an area. Let the occurrence of the generating mechanism be represented by a Poisson process, $\{N_1(t)\}$, with rate λ . (See Figure 1, in which $\{\tau_i\}$ denotes the arrival epochs of the generating mechanisms). Each generating mechanism triggers a random number ν of rainfall events with density p_k , $k = 1, 2, \dots$, and the probability generating function (pgf) $\Pi(u)$,

$$\Pi(u) = \sum_k u^k p_k \quad |u| < 1 \quad (17)$$

Assume that the time of occurrence of these events relative to the occurrence of the generating mechanism is governed by a sequence of iid random variables, $\{T_i\}$, with some density function $f(t)$ (see Figure 1). Also assume that the random process $\{N_1(t)\}$, the random variable ν , and the sequence $\{T_i\}$

are independent. Let $N(t) = N(0, t)$ denote the number of rainfall events in $(0, t)$ and let $\phi_A(u)$ denote the pgf of $N(t)$. Then

$$\phi_A(u) = \exp \left\{ -\lambda \int_{-\infty}^t (1 - \Pi[1 - (1 - u)q_A(x)]) dx \right\} \quad (18)$$

$$q_A(x) = \int_0^t f(s - x) ds$$

It may be noted that $q_A(x)$ represents the probability that a rainfall event generated by a mechanism located at x occurs within the interval $(0, t)$. Kavvas and Delleur [1975] also derived an expression for the joint pgf of $N(t_1)$ and $N(t_2)$, $t_1 < t_2$, the number of rainfall events in $(0, t_1)$ and $(0, t_2)$, respectively. A comprehensive analysis of the rainfall process requires that the joint densities of $N(t_i) - N(t_{i-1})$, $i = 1, 2, \dots, n$, $t_1 < t_2 < \dots < t_n$ or their joint pgf be known [see, e.g., Kavvas and Delleur, 1975]. This is discussed systematically in part 2 of this paper via more general transform techniques. An application of the general transform for the N-S process derived in part 2 is given in part 3.

The above discussion brings out the main ideas that have been exploited in the hydrology literature for modeling the counting processes associated with the rainfall events. Thus far we have not considered other important features of temporal rainfall (e.g., the structure of rainfall depths, rainfall duration, etc.). These features will be considered now.

Rainfall depth and duration processes. The reader may recall from the earlier discussion that $\{Y_i\}$, each Y_i taking values 0 or 1, denotes the sequence of zero rainfall or nonzero rainfall periods over some discrete time scale (e.g., a day, an hour, etc). Suppose $Y_0 = 1$. Let $T_r^{(1)}$ be the random duration for which there is uninterrupted rainfall, i.e.,

$$T_r^{(1)} = \min \{n; Y_n = 0\} \quad (19)$$

Then it follows from the assumption that $\{Y_i\}$ is Markov chain (recall (5)), that

$$P(T_r^{(1)} = k) = p_1^{k-1} (1 - p_1) \quad k = 1, 2, \dots \quad (20)$$

This shows that $T_r^{(1)}$ has a geometric density. Similarly, if $T_b^{(1)}$ denotes the number of days without rain following $T_r^{(1)}$, then it also can be shown to have a geometric density

$$P(T_b^{(1)} = k) = p_0^{k-1} (1 - p_0) \quad k = 1, 2, \dots \quad (21)$$

One can also prove that the sequence of random variables $T_r^{(1)}, T_b^{(1)}, \dots$ so constructed are independent. These facts are well known in the theory of Markov chains, and the reader may consult Feller [1968] for details. Figure 2 shows this scheme schematically. This construction provides a procedure for investigating the rainfall process through the analysis of the random sequences $\{T_r^{(i)}\}$ and $\{T_b^{(i)}\}$, generally called the length of wet spells (storm duration) and the length of dry spells (time between storms). However, investigations of these

sequences are usually within a continuous time frame rather than in a discrete time frame.

Grayman and Eagleson [1971] constructed sequences $\{T_r^{(i)}\}$ and $\{T_b^{(i)}\}$ from rainfall records in the units of 10-minute increments. A storm duration T_r was defined as the duration of a sequence of rainfall bursts not separated by more than two hours. The time between storms was defined as the sequence of zero rainfall intervals greater than two hours that separate two consecutive rainfall bursts. They assumed that $\{T_r^{(i)}\}$ are mutually independent and are independent of $\{T_b^{(i)}\}$. A Weibull density was fitted to $\{T_r^{(i)}\}$ and also to $\{T_b^{(i)}\}$, the parameters of the two Weibull densities being different. More recently, Eagleson [1978] proposed the use of exponential densities with different parameters for the respective sequences of wet and dry spells defined above. This postulate was supported by some rainfall data from Boston, Massachusetts.

The postulates on $\{T_r^{(i)}\}$ and $\{T_b^{(i)}\}$ made by Eagleson [1978] are a consequence of the Markovian assumption on the continuous time analogue, $\{Y(t); 0 \leq t < \infty\}$, of the discrete time Markov chain $\{Y_i; i \geq 0\}$. In the continuous time version, the geometric densities given by (20) and (21) become exponential, say with parameters λ_1 and λ_2 , where $1/\lambda_2$ is the mean dry length and $1/\lambda_1$ is the mean wet length. The continuous parameter Markov chain $\{Y(t)\}$ in the context of rainfall has been explored by Crovelli [1971]. He gives the expression for the transition density $P(t)$ for this Markov chain. Other variants of the two-state Markov chain $\{Y(t)\}$ can also be introduced (e.g., a k state Markov chain where each state then refers to a storm with its rainfall magnitude falling into one of the k states). Crovelli [1971] introduced a four-state Markov chain. The states are classified as dry, a trace storm, a moderate storm, and an intense storm. He also introduced other schemes for reducing a process to approximations of a Markovian description when, for example, $\{T_r^{(i)}\}$ and $\{T_b^{(i)}\}$ are not exponentially distributed. However, we will not go into the details of these descriptions as they are not particularly illuminating from a physical viewpoint.

Importantly, it is possible to introduce a counting process description of a rainfall process in terms of the sequences of the lengths of wet and dry spells. This is similar to the process $\{N_n\}$ constructed from the sequences $\{Y_i\}$ by Gabriel [1959]. Specifically, define a sequence of random times η_n , $i \geq 1$, as

$$\begin{aligned} \eta_1 &= T_r^{(1)} + T_b^{(1)} \\ \eta_2 &= T_r^{(2)} + T_b^{(2)} \\ \eta_k &= T_r^{(k)} + T_b^{(k)} \end{aligned} \quad (22)$$

$$N(t) = \max \left\{ k; \sum_{i=1}^k \eta_i < t \right\}$$

$N(t)$ denotes the total number of wet spells with their epochs of origination falling within $(0, t)$. Then from the identity [Karlin and Taylor, 1975, p. 169]

$$P(N(t) = k) = P \left(\sum_{i=1}^k \eta_i < t \right) - P \left(\sum_{i=1}^{k+1} \eta_i < t \right) \quad (23)$$

the marginal distribution of $N(t)$ can be obtained. Even the finite dimensional joint distributions associated with the process $\{N(t)\}$ can be derived, although such derivations are generally unwieldy. If the sequences of variables $T_r^{(i)}$ and $T_b^{(i)}$ are independent and exponential, then η_n , $i \geq 2$ are not exponentially distributed. Hence $\{N(t)\}$ is not a Markov process.

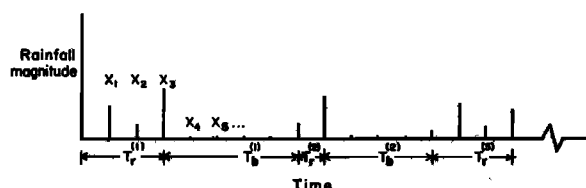


Fig. 2. A schematic depiction of an alternate sequence of wet and dry days.

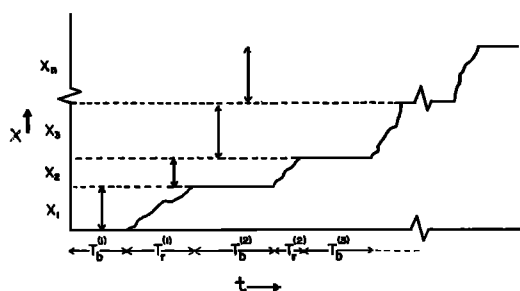


Fig. 3. A schematic depiction of the sample function of cumulative rainfall.

A physical simplification can be introduced into this scheme from the observation that generally $E[T_b] \gg E[T_s]$ (i.e., $\lambda_1 \gg \lambda_2$). It then follows that the density of η_i , $i \geq 1$, can be approximated by an exponential density with parameters λ_2 so that $\{N(t)\}$ becomes a Poisson process [see, e.g., Eagleson, 1978].

Rather than obtain a counting process from assumptions on the wet and dry duration sequences, Kavvas and Delleur [1975] compute the densities of wet and dry durations from the Neyman-Scott process description of the counting process $\{N(0, t)\}$. From an analytical as well as a physical viewpoint this idea seems promising. The reader should also see the second section of part 3.

Our discussion so far has been confined to the structure of storm durations and the time between storms. We now turn to a discussion of rainfall magnitudes. In the discrete time framework the reader may recall that the sequence $\{X_i; i \geq 1\}$, was introduced to represent the sequence of rainfall magnitudes. When the discretized time interval represents days, then the variables X_i have been assumed to be independent and to have a common exponential density [Todorovic and Woolhiser, 1976]. These assumptions are supported by some rainfall data from Austin, Texas. Other assumptions regarding the sequence $\{X_i\}$ have been introduced into the literature in the context of discretized time intervals other than a day. For example, Wang et al. [1981] assume the rainfall magnitudes averaged over every 4-hour time interval are independent and follow a common log normal density.

Since most of the models involving rainfall magnitudes are in the continuous time framework, we will now examine some of these. If $\{j(s); 0 \leq s < \infty\}$ denotes the rainfall intensity process, then the total magnitude of rainfall per storm having duration $T_r^{(i)}$ is given by

$$X_i = \int_0^{T_r^{(i)}} j(s) I_{T_r^{(i)}}(s) ds \quad i \geq 1 \quad (24)$$

where

$$\begin{aligned} I_{T_r^{(i)}}(s) &= 1 & s \in T_r^{(i)} \\ I_{T_r^{(i)}}(s) &= 0 & \text{otherwise} \end{aligned}$$

It is clear from (24) that in general X_i and $T_r^{(i)}$ for each i will be stochastically dependent. This dependence was empirically shown to exist for the rainfall data from Truro, Nova Scotia and St. Johnsbury, Vermont, by Grayman and Eagleson [1971]. They developed regression relations between X_i and $T_r^{(i)}$. Motivated by this analysis, Crovelli [1971] introduced a bivariate gamma density for the joint density of rainfall magnitude and duration. In (24), if one assumes that $j(s) = j_i$ is a constant, then $X_i = j_i T_r^{(i)}$. Using this relationship, along with the assumptions that j_i and $T_r^{(i)}$ are independent and follow

(different) exponential densities, Eagleson [1978] obtained an expression for the density of X_i in terms of the modified Bessel function of zero order. Interestingly enough, he showed that his data on rainfall magnitudes could be fitted well by this density as well as by a Gamma density, but not by an exponential density function. In contrast, the assumption of a common exponential density for the variables X_i has been exploited greatly by Todorovic and Yevjevich [1969] and Todorovic and Woolhiser [1976]. Before discussing their work we wish to remark that one obvious reason why an assumption can work well in one instance and not in another is the difference in the definitions of a storm. For example, Eagleson's [1978] definition of a storm is not the same as that of Todorovic and Yevjevich [1969].

An interesting idea was introduced by Todorovic and Yevjevich [1969] for analyzing the magnitudes. Figure 3 denotes a typical sample function (realization) of the cumulative rainfall process in which $\{X_i\}$ denote the total rainfall magnitudes obtained from successive storms. Define a counting process $\{M(x), 0 \leq x < \infty\}$ as

$$M(x) = \max \left\{ n; \sum_{i=1}^n X_i < x \right\} \quad (25)$$

Then $M(x)$ denotes the total number of storms whose cumulative rainfall yield is less than x . Under the assumption that the variables X_i are iid exponential, $\{M(x)\}$ becomes a Poisson process with parameter μ , say. Todorovic and Yevjevich [1969] assumed that $\{M(x)\}$ is a nonhomogeneous Poisson process with rate function $\mu(\cdot)$. In this case, the X_i lose their exponential character and exhibit dependence. Further details on the structure of X_i can be found in the work by Parzen [1967, p. 138].

Mathematically speaking, it is possible to define more general counting processes in the 'magnitude space'. For example, Todorovic and Woolhiser [1976] assume $\{M(x)\}$ to be a pure birth Markovian process and explore the particular case when it is a nonhomogeneous version of the Polya process. However, they do not motivate these assumptions from any physical (empirical) considerations. Therefore this postulate remains as yet physically obscure, although the basic idea may provide some insights on the structure of rainfall magnitudes.

In the next section we will briefly review some results that have been obtained from postulates on the rainfall counting process; magnitudes, and durations, as reviewed above.

Some results. It may be recalled from our earlier discussion that $\{j(s); s \geq 0\}$ is used to denote the rainfall intensity process. Although this process is of a fundamental importance in diverse hydrologic applications, most of the mathematical modeling of temporal rainfall is restricted to the process of cumulative rainfall, $\{X(t); t \geq 0\}$, where

$$X(t) = \int_0^t j(s) ds \quad t \geq 0 \quad (26)$$

In view of (22) it may be observed that $X(t)$ consists of the total rainfall due to $N(t)$, the number of complete storms, and possibly a partial contribution from the $(N(t) + 1)$ st storm. The appearance of this partial contribution complicates the analysis of $\{X(t); t \geq 0\}$. In order to introduce some simplification into this structure, one can define two auxiliary stochastic processes, $\{X_0(t)\}$ and $\{X_1(t)\}$, as follows [Todorovic and Woolhiser, 1976]:

$$X_0(t) = \sum_{i=1}^{N(t)} X_i \quad t \geq 0 \quad (27)$$

$$X_1(t) = \sum_{i=1}^{N(t)+1} X_i \quad t \geq 0 \quad (28)$$

It is clear from the above remarks that

$$X_0(t) \leq X(t) \leq X_1(t) \quad (29)$$

Now instead of investigating $\{X(t)\}$ directly, it becomes possible to investigate $\{X_0(t)\}$ as an approximation of $\{X(t)\}$. For the present we do not think it necessary to go into the details of this approximation. The interested reader may refer to *Todorovic and Woolhiser* [1976] as well as to some of Todorovic's earlier work referenced therein for further details. We will confine the ensuing discussion to considerations of $\{X_0(t)\}$. The analysis of $\{X_1(t)\}$ follows in a very similar manner.

If $F_0(x, t)$ denotes the marginal distribution function of $X_0(t)$, then in view of (27),

$$F_0(x, t) = P(N(t) = 0) + \sum_{n=1}^{\infty} P\left(\sum_{i=1}^n X_i < x, N(t) = n\right) \quad (30)$$

Under the assumption that the variables X_i are iid and independent of $\{N(t)\}$, and that $\{N(t)\}$ is a Poisson process, (30) represents the distribution function of a compound Poisson process. This result has been rederived in the hydrologic literature at different times by different researchers [see, e.g., *Le-Cam*, 1961; *Shane*, 1964; *Todorovic and Yevjevich*, 1969; *Duckstein et al.*, 1972; and *Eagleson*, 1978]. In the case of a compound Poisson process, (30) reduces to

$$F_0(x, t) = e^{-\Lambda(t)} + \sum_{n=1}^{\infty} F^{*n}(x) e^{-\Lambda(t)} \left[\frac{\Lambda(t)}{n!} \right]^n \quad (31)$$

where

$$F^{*n}(x) = P\left(\sum_{i=1}^n X_i < x\right) \quad (32)$$

and $\Lambda(t)$ is as it was defined earlier in (4). Specific assumptions on the variables X_i produce explicit and generally tractable forms for $F_0(x, t)$, the details of which need not detain us here.

From a physical viewpoint it can be seen that the above description of $X_0(t)$ gives camouflage to some important features of rainfall (i.e., the storm duration and the 'distribution' of intensity over this duration). These features are not only essential to a description of the rainfall intensity process, but also should play an important role in investigating the statistical behavior of, for example, the runoff process from a physical viewpoint. However, in some physical cases (e.g., in the case of summer thunderstorm rainfall in the Southwestern and the Southeastern United States) it may be reasonable to assume that rainfall occurs instantaneously in clumps of size X_n , $i \geq 1$, as implied by the above description of $X_0(t)$.

Some results beyond the compound Poisson description have been derived from (30) by *Todorovic and Woolhiser* [1976]. In view of the counting process $\{M(x); x > 0\}$ defined by (25), one can write (30) as

$$F_0(x, t) = P(N(t) = 0) + \sum_{n=1}^{\infty} \sum_{j=n}^{\infty} P(M(x) = j, N(t) = n) \quad (33)$$

Under the assumptions that $\{M(x)\}$ and $\{N(t)\}$ are independent and that each is governed by a nonhomogeneous version of the Polya process, specific expressions for $F_0(x, t)$ have been obtained by *Todorovic and Woolhiser* [1976]. They also consider the discrete time version of cumulative rainfall process, S_n ,

$$S_n = \sum_{i=1}^{N_n} X_i^+ \quad (34)$$

where the variables $X_1^+, X_2^+, \dots$ are iid with a common distribution given by the conditional distribution of X_i given that $\{X_i > 0\}$. It is important to note that in view of this assumption, the dependence in the sequence $X_1, X_2, \dots$ is only between the occurrence of wet and dry events and not between the rainfall magnitudes during wet periods. *Todorovic and Woolhiser* [1976] use (34) to derive an expression for the distribution of S_n , say $F(x, n)$. The derivation also assumes that N_n is given by (6).

In the results quoted above attention is essentially focused on a derivation of the one dimensional (marginal) distribution of $\{X_0(t)\}$. Since a stochastic process is determined by all its finite dimensional joint distribution functions, the above results give only an incomplete description of the process $\{X_0(t)\}$. An exception to this situation is represented by the compound Poisson process. Since the increments of this process are stochastically independent [*Feller*, 1968] its finite dimensional joint distributions can be written down immediately once its marginal distribution function $F_0(x, t)$ is known. In addition to the cumulative rainfall process, two other processes derived from rainfall have been investigated. These we will now review.

Let $X_0^*(t)$ denote the maximal rainfall per rainfall storm in an interval $(0, t)$. Then

$$X_0^*(t) = \max_{1 \leq i \leq N(t)} X_i \quad (35)$$

Under the assumption that the variables are iid and independent of $\{N(t)\}$, it follows that

$$P(X_0^*(t) < x) = F_0^*(x, t) = \sum_{n=0}^{\infty} F^n(x) P(N(t) = n) \quad (36)$$

where $F(x)$ denotes the common distribution function of X_n , $i = 1, 2, \dots$. This result was applied to thunderstorm rainfall in the southwestern United States by *Fogel and Duckstein* [1969]. Its analogue for the case of a rainfall description over discretized time intervals (days) has been investigated by *Todorovic and Woolhiser* [1976]. In this case, if X_n^* denotes the maximal rainfall in n days, then

$$Z_n^* = \max_{1 \leq i \leq N_n} X_i \quad (37)$$

$$P(X_n^* < x) = \sum_{j=0}^{\infty} F^j(x) P(N_n = j) \quad (38)$$

Equation (38) is derived from the assumptions that the variables X_i are iid and are independent of $\{N_n\}$. The two examples investigated by *Todorovic and Woolhiser* [1976] are for the two cases when N_n is given by (2) and (6). Once again the expressions given by (36) and (38) are only for the marginal distributions of the stochastic processes $\{X_0^*(t); t > 0\}$ and $\{X_n^*; n \geq 0\}$.

Investigations of droughts have motivated some analyses on the structure of the maximal dry period within an interval $(0, t)$ or $(0, n)$. In the case of a continuous time description, Gupta and Duckstein [1975] have formulated this problem as that of determining the maximal interarrival time of a Poisson process within some interval $(0, t)$. Clearly, this formulation is based on the assumptions that $\{N(t)\}$ is a Poisson process and that rainfall takes place instantaneously. The distribution function of the maximal interarrival time in a homogeneous Poisson process is well known in the probability literature [Feller, 1971]. Its extension to the nonhomogeneous case was derived by Gupta and Duckstein [1975]. The derived expressions are somewhat unwieldy and will not be reviewed here. A similar analysis was given by Millan [1972] for the process $\{Y_t\}$ defined in (1) under the assumption that it is a Markov chain. The probability distribution of the maximum uninterrupted sequence of zeros within an interval $(0, n)$ was computed. Again, the resulting expression is unwieldy and will not be given here.

In the analyses reported above, expressions for the mean and variance of the random variables under investigation have also been derived. Some of these derivations, for example, in the case of a compound Poisson process, are standard in the probability literature [see, e.g., Karlin and Taylor, 1975]. Some further results in addition to those reviewed above have been obtained by Todorovic and Woolhiser [1974]. For example, they prove a central limit theorem (CLT) for S_n as defined by (34). However we will not stop to review these results and will go on to review the models of the space-time rainfall process.

THE STRUCTURE OF RAINFALL IN SPACE AND TIME

Models of rainfall in space and time can be grouped into three broad categories. The first category consists of the numerical modeling of space-time rainfall on the basis of the principles of fluid mechanics, thermodynamics, etc. [see, e.g., Weinstein and Davis, 1968; Young, 1974]. The second category includes the numerical simulation of rainfall on the basis of certain empirical regularities which space-time rainfall has been observed to exhibit [see, e.g., Grayman and Eagleson, 1971; Sorman and Wallace, 1972; Corotis, 1976; Amorocho and Wu, 1977]. The third category includes analytical modeling of rainfall as a random field [LeCam, 1961; Bras and Rodriguez-Iturbe, 1976; Gupta and Waymire, 1979].

In delineating the statistical structure of space-time rainfall, it would of course be ideal if the physical laws governing the occurrence and distribution of rainfall could be directly incorporated into its statistical description. However, to the best of our information, neither has such a scheme been carried out, nor does it seem imminent in the near future. For this reason the approaches under the second and the third categories remain pretty much isolated from the approach in the first category. Keeping in mind that our present objective is to review only the statistical descriptions of space-time rainfall, it seems that a review of the approaches under the first category is outside the scope of this article.

Many of the approaches under the second and the third category have been motivated by the observed regularities in the evolution of space-time rainfall from diverse storm types [Houze, 1969; Austin and Houze, 1972; Wu, 1973]. For this reason we begin this section with a summary of certain regularities in the structure of the space-time rainfall which have

been identified through numerous observational studies. The following section is essentially extracted from Gupta and Waymire [1979].

A summary of observed regularities (clustering) in subsynoptic space-time rainfall. Starting with the pioneering work of Byers and Braham [1949], a number of investigators have gradually synthesized an empirical space-time description of subsynoptic rainfall (see, e.g., Houze [1969], Austin and Houze [1972], Sorman and Wallace [1972], Stall and Huff [1971], Wu [1973], Amorocho and Wu [1977], etc.). Systematic observations on diverse storm types (e.g., cyclonic storms in the New England region and in northern California, air mass thunderstorms and frontal and squall line thunderstorms in the Southeast, the Midwest, and the Southwest United States, etc.) indicate the consistent occurrence of certain subsynoptic features with similar characteristics and behavior. These subsynoptic features have been reported in detail by Austin and Houze [1972].

Precipitation areas have been categorized according to their horizontal extent. Areas greater than about 10^4 km² are referred to as synoptic areas. For example, synoptic areas are associated with cyclonic storms, while, on the other hand, such large-scale extent would not be found in the frontal passage or air mass thunderstorm. Synoptic areas generally have a lifetime of one to several days. The largest subsynoptic precipitation areas contained within a synoptic area are called large mesoscale areas (LMSA). For cyclonic storms in the northern California and New England regions the LMSA generally appear as elongated bands, and consequently, they have also been called 'bands' [Amorocho and Wu, 1977]. The LMSA, ranging in extent from 10^3 to 10^4 km², build and dissipate within a synoptic area. Their life span has been observed to be within the order of several hours, and their number within a synoptic area has been observed to range anywhere from one to six. The LMSA move in relation to a moving synoptic region, and their motion can at times be faster than the motion of the synoptic region. The precipitation intensity inside an LMSA is always higher than the region surrounding it. This high intensity produces a region of high reflectivity (relative to its immediate environment) on the plan position indicator (PPI) of a radar which enables an observer to identify the presence and extent of an LMSA.

Each LMSA contains identifiable regions of cumulus convective precipitation, commonly known as 'convective cells'. The convective cells, ranging from 10 to 30 km² in extent depending upon the storm type, build and dissipate within an LMSA and generally occur in clusters. The life span of a cell is of the order of several minutes to about half an hour and can be divided into three stages of development. These are the cumulus stage, the mature stage, and the dissipating stage [Petterssen, 1956; Amorocho and Wu, 1977]. These cells undergo motion relative to the motion of the LMSA and the rainfall intensity within a cell is always higher than the intensity in the region surrounding it. A likely explanation for the phenomenon of clustering has been succinctly stated by Petterssen [1956, p. 161] in the case of thunderstorm rainfall:

Observations show that there is a distinct tendency for new cells to form in the immediate vicinity of existing cells. The most likely explanation of this tendency for clustering is, perhaps, that the downdraft is compensated for by upward motion in the environment and that the cold air that spreads out from the downdraft of one cell gives an upward impulse to the neighboring warm air, with the result that a new convective cell is initiated.

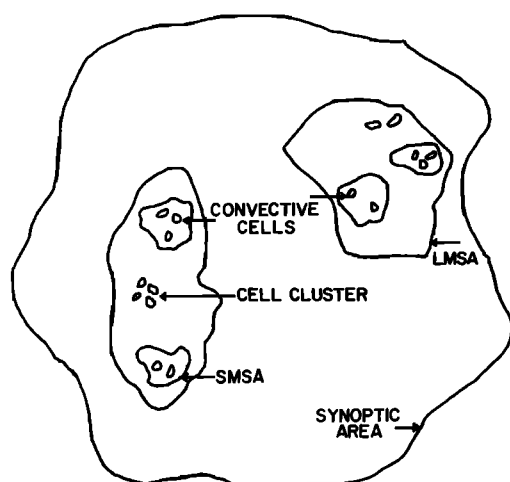


Fig. 4. A schematic depiction of subsynoptic rainfall features.

Observations of *Austin and Houze* [1972] indicate the possibility that cell clusters are not the only subregions in which cells may be found. Their observations on certain storms indicate the occurrence of small mesoscale precipitation areas (SMSA), ranging from 10^2 to 10^3 km² in extent, within an LMSA. Each SMSA in turn contains rainfall cells. The SMSA build and dissipate within an LMSA with an average life span of a few hours. *Austin and Houze* [1972, p. 933] state

One of the most notable features of the small mesoscale areas is an apparent close relationship with the convective cells, shown by the facts that every SMSA contained at least one cell and that their motions were similar to those of cells within them.

The distinguishing feature between an SMSA containing cells and a cluster of cells is that in an SMSA the rainfall intensity is higher than that in the LMSA but lower than that in the cells, while in a cell cluster the rainfall intensity in the region surrounding each cell is simply the intensity of the LMSA. Although the horizontal extents of a cell cluster and an SMSA containing cells are similar, in some instances it has been observed that the percentage of rainfall in the cells within an SMSA is small in comparison with the regions of SMSA surrounding these cells. Further discussion of the SMSA and the possible physical mechanisms responsible for their occurrence can be found in the work of *Austin and Houze* [1972]. The organization of the different subsynoptic entities is schematically depicted in Figure 4.

The above discussion of the general features of subsynoptic rainfall was largely derived from observations on cyclonic storms; however, similar characteristics have been exhibited by many other storm types covering a wide range of synoptic and seasonal situations. For example, while the air mass thunderstorms do not exhibit synoptic and LMSA regions, they do occur in the form of isolated convective cells or clusters of cells. Similarly, although the precipitation associated with frontal bands is not at the synoptic scale level, frontal bands do cover an area ascribed to an LMSA containing several SMSA within it. Since an LMSA depicts the general structure common to a variety of storm types, the quantification of subsynoptic rainfall has been taken to be tantamount to quantifying the rainfall field from an LMSA. Furthermore, in hydrologic studies where precipitation inputs in space-time are required, the spatial regions covered by large mesoscale areas are generally the regions of greatest significance.

Mathematical structure of space-time rainfall. Keeping in mind the above description of the space-time rainfall, a mathematical description amounts to describing the random field due to the passage of a synoptic scale disturbance over a given region of interest. The latter could naturally be called a storm [*Amorocho and Wu*, 1977]. In this connection the times between storms, $\{T_b^{(n)}\}$, have been defined in the same manner as was done for the temporal rainfall process. The storm durations $\{T_d^{(n)}\}$ also appear as analogs to the temporal rainfall [*Gupta*, 1973; *Amorocho and Wu*, 1977; *Bras and Rodriguez-Iturbe*, 1976]. The assumptions on these variables vary from one author to another. For example, *Amorocho and Wu* [1977] assume the variables $T_b^{(n)}$ to have a Weibull density whereas *Bras and Rodriguez-Iturbe* [1976] take it to be an exponentially distributed sequence. Given that a synoptic disturbance has originated, the main problem in modeling rainfall is to describe the mathematical structure of the random field (of rainfall intensity) resulting from this disturbance. In this connection the reader may note that the structure of the random field implied by simulation models is not always simple to identify. For this reason we will not go into the details of the simulation schemes that have been introduced into the literature (such as those by *Sorman and Wallace* [1972] and *Amorocho and Wu* [1977], where the underlying random field has not been explicitly identified). In contradistinction to these we will review, for example, the approach of *Bras and Rodriguez-Iturbe* [1976]. Although their model is also a simulation scheme, it explicitly identifies the mathematical structure of the random field which it simulates.

The first detailed analytical study of the precipitation field is contained in the pioneering work of *LeCam* [1961]. Although *LeCam* did not give a detailed construction of the rainfall field in order to obtain explicit results, he nonetheless laid the foundations of a mathematical framework which could accommodate most of those mathematical operations which arise out of attempts to represent rainfall in space and time.

Gupta [1973] conceptualized the structure of subsynoptic rainfall through the occurrence of thunderstorm cells and assumed that the random field denoting the number of cells in some region B , conditioned on the duration of a storm, is Poisson. This assumption ignores the empirically observed spatial clustering of cells as well as their motions. More importantly, this formulation does not deal with the rainfall intensity field. It directly models the cumulative rainfall process (in time). Consequently the spatial evolution of rainfall is accounted for very grossly in this approach. Expressions are derived for the mean and variance of the cumulative rainfall under the assumption that the rainfall magnitudes per storm are stochastically dependent on the counting process denoting the number of storms in a certain time interval over a fixed region B . Mathematically speaking, this assumption is analogous to the assumption that the variables X_i in (27) depend on η_i , $i \geq 1$. Since the variables η_i defined by (22), determine the $\{N(t)\}$ process, *Gupta's* results represent a particular relaxation of the assumption generally made in the temporal rainfall modeling that the X_i and the η_i are independent.

A detailed formulation of the stochastic rainfall intensity field was recently given by *Gupta and Waymire* [1979]. This formulation is derived from the physical description of the evolution of subsynoptic rainfall as summarized earlier. Four major assumptions are made on the components of a LMSA and on the cells originating and dissipating within a LMSA.

1. $L(\cdot)$ is a Poisson random field on B_0 , the region occupied by a mesoscale area (LMSA). The intensity parameter of the Poisson field is $\lambda_0 > 0$.

2. $N_x(\cdot)$ is a Poisson random field on C_x , where

$$C_x = \{(\tau, y) \in \mathcal{R}^2 : \|y - x - \bar{U}_b \tau\| < K\} \\ \mathcal{R} = (0, \infty) \times \mathbb{R}^2 \quad (39)$$

$\bar{U}_b$ denotes the LMSA velocity and K denotes the radius of a cluster potential. Physically, a cluster potential is a region within B_0 which produces a cluster of cells at different time points. Equation (39) states that the occurrence of these cells in time, within a cluster potential, is Poissonian. If $A \subset C_x$, then $E[N_x(A)] = \Lambda(x, A)$.

3. The random fields in the sequence $\{N_x\}$ are mutually independent as well as independent of $L(\cdot)$.

4. A function $f(a, r)$, denoting the rainfall intensity for a fixed cell, is defined to satisfy (1) $f(a, r)$, a decreasing function of r for each a and (2) $f(a, r) = 0$ if $r > R(a)$ and/or $a > T$. The spatial extent of a cell at age a is denoted by $R(a)$ and T denotes the cell duration. Let $\bar{U}_c$ denote the cell velocity relative to the velocity of the LMSA.

The rainfall intensity $j(t, z)$ at some point z at time t is simply taken to be the superposition of all intensities from the various cells born and dissipated within a moving LMSA and moving with a common velocity $\bar{U}_c$. Mathematically this can be expressed as

$$j(t, z) = \int_{B_0} \left\{ \int_{\mathcal{R}} f(t - \tau, \|z - (y + \bar{U}(t - \tau))\|) \cdot dN_x(\tau, y) \right\} dL(x) \quad t > 0 \quad z \in \mathbb{R}^2 \quad (40)$$

where $\bar{U} = \bar{U}_b + \bar{U}_c$. Equation (40) gives a mathematical representation of the random field $\{j(t, z)\}$ representing rainfall intensity. Gupta and Waymire [1979] define two auxiliary stochastic fields in order to determine the probability structure of $\{j(t, z)\}$. These auxiliary fields, suggested by (40), are $\{N(A)\}$ and $\{M_s'(x, G)\}$. The first field $\{N(A)\}$ represents the number of birth center points of the convective cells within a LMSA as exhibiting spatial clustering. The second field $\{M_s'(x, G)\}$ represents the number of cells born in an interval (s, t) within a moving set G of $\mathbb{R}^2$ supported by their cluster potentials with center at $x \in B_0$. Expressions for the finite dimensional characteristic function of the field $\{N(A)\}$ and an expression for $E[N(A_1)N(A_2)]$ for disjoint A_1 and A_2 are derived. Because of spatial clustering, $N(A_1)$ and $N(A_2)$ exhibit dependence. From the field $\{M_s'(G)\}$, the stochastic process $M(t)$ is defined:

$$M(t) = M_{t-T}'(G) \quad (41)$$

It is shown that although the marginal density of $M(t)$ is Poisson, it is not a Poisson process. It is shown to exhibit a T step dependence in that $M(t)$ and $M(s)$ are dependent for $|s - t| < T$ and are independent if $|s - t| > T$. Lastly, expressions for the mean and variance of $j(t, z)$ and for the characteristic function of its marginal density are derived. Of course this represents only a preliminary study, and apparently much work remains to be done to further understand the structure of $\{j(t, z)\}$.

The last paper to be reviewed on space-time rainfall is that of Bras and Rodriguez-Iturbe [1976]. This approach also quantifies the rainfall intensity field $\{j(t, z)\}$ due to the passage of a storm. Their basic assumption is that $\{j(t, z)\}$ is a non-

stationary Gaussian random field. The nonstationarity is introduced through the temporal dependence of the mean $\{J(t, z)\}$ and the variance $\sigma^2(t, z)$ on time. If $\{e(t, z)\}$ denotes a normalized Gaussian random field, then

$$j(t, z) = \bar{J}(t, z) + e(t, z)\sigma(t, z) \quad (42)$$

The specification of $\bar{J}(t, z)$ and $\sigma(t, z)$ is accomplished through the construction of a mean intensity versus rainfall duration curve $i_a(t)$ and a mean standard deviation versus duration curve $\sigma_a(t)$. This construction can be accomplished by the observed rainfall hydrograph at different points over a region. They define

$$\bar{J}(t, z) = i_a(t - z_1/U_1) \\ \sigma(t, z) = \sigma_a(t - z_1/U_1) \quad (43)$$

where U_1 is a unidirectional storm velocity along the z_1 direction, $z = (z_1, z_2, z_3)$. In order to specify the Gaussian field $\{j(t, z)\}$ completely, it is sufficient to define the covariance function, say, $r(t_1, x; t_2, z)$, of the normalized field $\{e(t, z)\}$ (see (42)). They postulate that $r(\cdot)$ satisfies Taylor's hypothesis on turbulence. According to this hypothesis,

$$r(t_1 + s, x; t_2 + s, z) \\ = r(t_1, x_1 + U_1 s, t_2, z_1 + U_1 s) = r(0, x_1 \\ + (t_1 + s)U_1; 0, z_1 + (t_2 + s)U_1) \quad (44)$$

Equation (44) states that the correlation at times t_1, t_2 is the same as the spatial correlation at time zero, provided the spatial coordinates at time 0 are translated by $U_1 t_1$ and $U_1 t_2$ in the direction of the storm motion. Their motivation for introducing Taylor's hypothesis comes from the empirical studies of Zawadzki [1973]. The observation that the rainfall intensity covariance satisfies Taylor's hypothesis is indeed interesting. In view of (44) it is sufficient to specify the spatial correlation at time zero, say $\rho(x, z)$. Bras and Rodriguez-Iturbe [1976] assume that $\rho(x, z) = \rho(\|x - z\|)$, i.e., the random field is isotropic. Since the structure of the spatial correlation of rainfall intensities has not been investigated to any great extent, they propose three functional forms as examples. Once the covariance is known, it is possible to simulate the Gaussian field. A detailed algorithm for such a simulation, developed by Mejia and Rodriguez-Iturbe [1974], was employed for this purpose. It is beyond the intent of this paper to go into the details of this simulation algorithm.

A PERSPECTIVE ON THE DIRECTIONS FOR FURTHER RESEARCH

In the above review the main mathematical ideas that have been introduced into the literature for modeling rainfall were outlined. It can be seen from this discussion that a considerable diversity exists among these different models. One must wonder why this diversity is so prevalent. Is it solely because of the differences among different storm types in different parts of the globe? Undoubtedly this is part of the reason. But then what about the regularities that are observed? What type of a common mathematical framework can accommodate the observations? An answer to these questions obviously must await a considerable amount of further research. An immediate concern is whether or not it is possible to single out some specific directions which will unify the existing theory. There are two issues that come to mind in this context. The first issue concerns the synthesis of the empirical evidence that has been acquired through the past research efforts. The second issue

concerns a suitable mathematical framework which can accommodate the modeling of the type of regularities suggested by the empirical evidence. In the ensuing discussion we will attempt to synthesize some regularities in the structure of the diverse rainfall types. Through this discussion a framework emerges which cannot be quantified naturally using the more popular mathematical techniques (e.g., the theory of Markov chains).

In connection with the rainfall storm counting process, the binomial and the Poisson models give only a crude approximation to the modeling of daily rainfall events for the thunderstorm rainfall in the southwestern United States [Smith and Schreiber, 1973]. The situation becomes worse for smaller time scales (e.g., hours). If one introduces the non-Markovian counting process $\{N_n\}$ from a Markovian assumption on the sequence of wet-dry days [Gabriel and Neumann, 1962], some improvement over the binomial model results. But then this structure does not account for the dependence in the storm events in many other parts [Wiser, 1965]. Interestingly enough, Wiser compared an empirical plot of the exceedence probabilities of the length of dry days for San Francisco, California, and Greensboro, North Carolina, with the theoretical results from four counting processes including the binomial model and the Gabriel and Neumann [1962] non-Markovian model for $\{N_n\}$. He showed that some variants of the Polya model compare much better with the empirical results than either the Gabriel-Neumann model, the binomial model, or the Polya model itself. A similar conclusion was reached by Kavvas and Delleur [1975]. Kavvas and Delleur showed that the exceedence probabilities of a Neyman-Scott process for the daily rainfall fits equally well or better with the empirical data than that of a Poisson model. The mathematical scheme of Kavvas and Delleur falls into the realm of the point process theory (reviewed in part 2) and represents a break from the more traditional description of the observed rainfall features. In particular, the data analysis furnished by these authors can be extremely useful with regard to the future understanding of cluster structures in rainfall. What can be inferred from these observations? In view of the space-time description of rainfall given earlier, the following picture seems to emerge for the counting process.

Suppose that a storm front moves through a region. Then a given point in the region experiences rainfall from one or more convective cells, as well as from the background rainfall intensity within a SMSA. These are recorded as sequences of bursts, separated by some time period, which could denote the precipitation from clusters of cells from different SMSA's. The Neyman-Scott model, as a Poisson-cluster process, (this term will be defined in detail in part 2) incorporates this scheme directly. But then what about the Polya process [Todorovic and Woolhiser, 1976], or other variants of it introduced by Wiser [1965]? We will show in part 3 that the structure of the Polya process is not at all like that of the cluster process. On physical grounds, therefore, the Polya process appears somewhat ad hoc as a model for the occurrence of rainfall events. This observation is consistent with the empirical findings of Wiser [1965] as well as with the statement made by Todorovic and Woolhiser [1976] that the 'true' counting process should be somewhere between a Poisson model and a Polya model. The notion of a cluster process will be discussed in detail in part 2 [see, e.g., the section on Moyal's cluster process], and its suitability in modeling rainfall counting processes will be demonstrated in part 3.

With regard to modeling storm durations and magnitudes within the context of a cluster scheme, we can infer the following. Here one looks for the interarrival times of the origin of the clusters. The spacing of the cluster members (say storm events) undoubtedly will produce some random duration for the cluster, but it is not necessary to model it directly. However, it is necessary to verify the distribution of the durations resulting through a clustering scheme by comparison with empirical observations. Similarly, the space-time rainfall structure can offer some guidance with regard to modeling the magnitude of rainfall per storm event. For example, Grace and Eagleson [1966] define a rainfall storm as the rainfall recorded from 10-minute time increment intervals separated by less than two hours. They further observed that the rainfall magnitudes for events separated by more than two hours are eventually uncorrelated. This is interesting because the average life span of a cluster of cells of an SMSA is about two hours. Hence the scheme proposed by Grace and Eagleson suggests that their rainfall event could have consisted of all the rainfall recorded from a single SMSA and the cells within it. If one envisions that the rainfall magnitudes due to different SMSA's are independent, then it would be consistent with the empirical observations of Grace and Eagleson. However, if one defines a storm event as the rainfall resulting from bursts separated by less than two hours or so, the independence among rainfall magnitudes would not be valid. In that case some dependence scheme would be necessary. It is clear that more research is needed to identify these dependence structures. A very fruitful avenue into this realm would be to extract these dependence or independence structures in temporal rainfall magnitudes from a space-time description of rainfall.

The discussion on the space-time modeling of rainfall clearly suggests that a great deal of research needs to be done in this area. The role of the notion of a random field in this context is noteworthy. But, beyond that, the entire area seems to be wide open. For example, the Gaussian random field structure, exploited by Bras and Rodriguez-Iturbe [1976] has not been shown to result from a physical description of rainfall [see, e.g., Amorochio and Wu, 1977]. The framework proposed by Gupta and Waymire [1979] is only in its infancy, and analytic tools have not yet been applied to that framework so that quantitative details can be extracted from it and verified against empirical observations. In extracting quantitative results from this or any other similar framework, the role of the theory of point processes seems invaluable. It will be shown in Part 3 that the random field constructed by Gupta and Waymire has the Poisson clustering process (field) as its major building block. The representation (40) can be easily understood through a so-called smoothing operation on this Poisson cluster field [see the third section of part 2 and the fourth section of part 3 for details].

Although the above discussion centers around rainfall phenomenon, one cannot ignore the role of rainfall modeling in investigating other hydrologic processes which are driven by rainfall. Without going into the details of the state of the art of this area, one can infer from the recent review article of Klemesš [1978] that in the statistical modeling of streamflows from a physical viewpoint the rainfall descriptions are introduced in a very simplistic manner. In the derivation of the statistical behavior of streamflows from a physical viewpoint the source of randomness is normally taken to be the rainfall process. Yet the descriptions of rainfall generally involve very

strong assumptions (e.g., that the sequence of rainfall magnitudes is iid). These assumptions are dictated more by the choice of mathematical tools (e.g., an autoregressive, a moving average, or an ARMA model) than by the physical description of rainfall. It will be shown in part 3 that a 'coupling' of a realistic stochastic description of rainfall with that of a rainfall-runoff transformation leads to the point process framework. This, by the way, was first realized by LeCam [1961].

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